b. Let K be the splitting field of the irreducible polynomial $f(x)$ of degree 5 over $\mathbb{Q}$. Because each element of $G(K/\mathbb{Q})$ corresponds to a permutation of the five zeros of $f(x)$ in K, and multiplication is function composition for both automorphisms and permutations, we can view $G(K/\mathbb{Q})$ as a subgroup of S_5. Now $|G(K/\mathbb{Q})|$ is divisible by 5, for a zero α of $f(x)$ generates $\mathbb{Q}(\alpha) < K$ of degree 5 over $\mathbb{Q}$, and degrees of towers are multiplicative. By Sylow theory, a group of order divisible by 5 contains an element of order 5, which we can view as a cycle of length 5 in S_5.

The automorphism σ of $\mathbb{C}$ where $\sigma(a+bi) = a-bi$ induces an automorphism of K, which must carry one complex root $a+bi$ of $f(x)$ into the other one $a-bi$ and leave the real roots of $f(x)$ fixed. Thus this automorphism of K is of order 2, so we can view it as a transposition in S_5.

c. We find that $f'(x) = 10x^4 - 20x^3 = x^3(10x-20)$, so $f'(x) > 0$ where $x > 2$ or $x < 0$, and $f'(x) < 0$ for $0 < x < 2$. Because $f(-1) = -2, f(0) = 5$, and $f(2) = -11$, we see that $f(x)$ has one real zero between -1 and 0, one beween 0 and 2, and one greater than 2. These are all the real zeros because $f(x)$ increases for $x > 2$ and for $x < -1$. Thus $f(x)$ has exactly three real zeros and exactly two complex zeros so the group of the polynomial is isomorphic to S_5 and the polynomial is not solvable by radicals.

APPENDIX: Matrix Algebra

1. $\begin{bmatrix} 2 & 1 \\ 2 & 7 \end{bmatrix}$ **2.** $\begin{bmatrix} 4+i & -3+i & 1 \\ 7-i & 1+2i & 2-i \end{bmatrix}$ **3.** $\begin{bmatrix} -3+2i & -1-4i \\ 2 & -i \\ 0 & -i \end{bmatrix}$

4. $\begin{bmatrix} 3 & 1 \\ 5 & 15 \end{bmatrix}$ **5.** $\begin{bmatrix} 5 & 16 & -3 \\ 0 & -18 & 24 \end{bmatrix}$ **6.** Undefined

7. $\begin{bmatrix} 1 & -i \\ 4-6i & -2-2i \end{bmatrix}$ **8.** $\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}^4 = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}^2 = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$

9. $\begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}^4 = \begin{bmatrix} 2 & -2i \\ 2i & 2 \end{bmatrix}^2 = \begin{bmatrix} 8 & -8i \\ 8i & 8 \end{bmatrix}$

10. $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 2 & 2 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix}$

11. $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ **12.** $\begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ **13.** (3)(2)(-8) = -48

14. Given that A^{-1} and B^{-1} exist, the associative property for matrix multiplication yields $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n$ so AB is invertible. Similarly, we compute $(A^{-1}B^{-1})(BA) = A^{-1}(B^{-1}B)A = A^{-1}I_nA = A^{-1}A = I_n$ so BA is invertible also.